

Σ_0 — The Collapse Certificate

A computable stability certificate for convergence dynamics, and an honest account of why
an ungrounded self-improving system tends to collapse or diverge

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Abstract

We give a *computable* certificate for contraction in a dissipative nonlinear system, and use it to make a precise, falsifiable claim about recursively self-improving systems. Splitting the local Jacobian’s symmetric part into active and near-null subspaces yields a Lyapunov function $V(x) = \frac{1}{2} \|P_M x\|^2$ whose decay is *guaranteed* when the active spectral abscissa is negative *and* the active subspace is invariant under the full Jacobian (e.g. A normal). We are explicit about that hypothesis: the general non-normal case is not covered by the energy argument, and the authoritative test is the full spectrum $\max \operatorname{Re} \lambda(A) < 0$. For the non-normal regime we add two sufficient, provable contraction gates — a numerical-range gate and a Lyapunov/similarity-metric gate (§4) — that strictly widen the certified region, with provably-correct transient bounds. We pair the theorem with three control constructs: a collapse operator Σ_0 (a heuristic trigger plus projection onto the null manifold, the “42-state”); an anti-collapse operator Σ_0^{-1} , for which we prove a closed-form *one-step anisotropy-lift lemma* (§6) and report a 900-run forced-collapse sweep (100% prevented); and a Kalman innovation-surprise “canary.” Read together, the construction is a small, honest warning: absent external grounding, the linearized dynamics admit only collapse or divergence, and grounding is the safety mechanism — the control-theoretic shadow of *model collapse*. Theorem 1 and Lemma L2 are machine-checked (`tests/test_cio_sde.py`, 34 passing; the gates in `tests/test_stability_gates.py`, 14 passing). Components that are heuristic are labelled as such; the AI-safety reading is the keeper and stands on the published model-collapse literature.

Per-section status (honest by construction). §3 theorem: *proven under an explicit hypothesis, machine-checked for the symmetric case.* §4 non-normal gates: *proven sufficient (not necessary), machine-checked.* §5 trigger: *operational definition, not derived.* §6–7 anti-collapse: *L2 lift proven closed-form; full sufficiency theorem open; suppression measured over 900 runs.* §8 canary: *eigenvalue readout retired; innovation-surprise signal implemented.* §10 demonstration: *reproducible (scripts committed, real 2678-turn run).* §11 AI-safety: *the keeper, on the model-collapse literature.*

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1 Introduction and contributions

Self-improving systems — optimizers, agents, models trained on their own output — share a failure geometry. With no contact with an external reality, the dynamics either *freeze* onto a single self-consistent state or *run away*. Practitioners observe the first as *model collapse* [12, 13, 14] and the second as instability or reward-hacking [15, 16]. This paper gives the control-theoretic skeleton of that observation, together with a computable certificate one can run on a local Jacobian.

Contributions.

1. A **computable contraction certificate** (Theorem 1): a Lyapunov split on the symmetric part of the Jacobian, with an exponential-decay guarantee under an explicit invariance hypothesis, machine-checked for the symmetric case.
2. **Two provable region-wideners** for non-normal A (§4): a numerical-range (matrix-measure) gate and a Lyapunov similarity-metric gate, each strictly wider than the small-gain bound, with provably-correct transient constants (Crouzeix–Palencia $1 + \sqrt{2}$).
3. A **closed-form one-step anisotropy-lift lemma** (Lemma 1, §6): one aligned anti-collapse bump of magnitude $\geq \Delta$ provably breaks the collapse trigger’s flatness condition. This is the deductive core toward a non-collapse theorem; it is unconditionally true and machine-checked.
4. An **empirical anti-collapse result** (§7): over 900 trials that collapse without protection, Σ_0^{-1} suppresses collapse in 100%.

5. A **reproducible demonstration** on a real 2678-turn conversation log (§10) showing ungrounded recursion settling onto a low-dimensional fixed point.
6. An **AI-safety reading** (§11) that ties the certificate to the model-collapse literature and states the “grounding is the safety mechanism” principle precisely, with honest scope.

Stance on honesty. Every claim below carries a status tag (*proven, measured, heuristic*). This is not decoration: a stability certificate that overclaims is worse than none, because it is trusted. Where the proof covers only a special case, we say so; where a number is illustrative rather than pinned by a test, we say so.

2 The object

We study a dissipative nonlinear system

$$\dot{x} = f(x, u, \theta), \quad x \in \mathbb{R}^n, \quad (1)$$

with internal state x (for our running example, a conversation’s encoded state), control / persistent-excitation input u , and slowly-varying parameters θ . Linearizing along a trajectory x^* gives the local Jacobian

$$\delta\dot{x} = A \delta x, \quad A = \left. \frac{\partial f}{\partial x} \right|_{x^*}, \quad (2)$$

and everything below reasons about the eigenstructure of A and its symmetric part $A_s = \frac{1}{2}(A + A^\top)$. The skew part $A_k = \frac{1}{2}(A - A^\top)$ carries rotation and is *not* seen by A_s ; the gap between the two is exactly what makes the general case harder than the symmetric one.

3 The collapse-guarantee theorem

Split the state space using A_s into the *near-null* subspace $N = \text{span}\{v_i : |\lambda_i(A_s)| < \varepsilon\}$ and its orthogonal complement, the *active* subspace M with projector P_M . Define the Lyapunov function on the active modes only,

$$V(x) = \frac{1}{2} \|P_M x\|^2, \quad (3)$$

and let the active spectral abscissa be $\alpha = \max\{\lambda_i(A_s) : v_i \in M\}$.

Theorem 1 (Contraction on the active subspace). *Assume $\alpha < 0$ **and** the active subspace M is A -invariant, i.e. $P_M A P_N = 0$ (equivalently: A is normal, $A = A_s$, or A commutes with P_M). Then*

$$\dot{V} \leq 2\alpha V \implies \|P_M x(t)\| \leq \|P_M x(0)\| e^{\alpha t}. \quad (4)$$

The active modes decay exponentially at rate $|\alpha|$ and the trajectory contracts onto the invariant null manifold N : under this hypothesis, collapse is guaranteed. If $\alpha \geq 0$, some active mode is non-contracting and collapse is not guaranteed.

Proof. Differentiating V along the flow,

$$\dot{V} = (P_M x)^\top P_M A x = (P_M x)^\top A_s (P_M x) + \underbrace{(P_M x)^\top A (P_N x)}_{\text{cross term}}.$$

The first term is $\leq 2\alpha V$ by the definition of α . The cross term vanishes exactly when M is A -invariant ($P_M A P_N = 0$), which holds automatically if A is normal. Under that hypothesis $\dot{V} \leq 2\alpha V$, and Grönwall’s inequality gives the exponential bound. $\square$

3.1 Why the hypothesis is required

For a general non-normal A the skew part A_k couples the active and null components, the cross term need not vanish, and the simple energy bound can fail.

Counterexample. For $A = \begin{pmatrix} -1 & 3 \\ -3 & 0 \end{pmatrix}$ the active abscissa is $\alpha = -1 < 0$, yet a direct scan finds $\max \dot{V}/V \approx +8.8 \times 10^4 \gg 2\alpha = -2$, and integrating from $x_0 = (0.3, 1.0)$ makes V *grow* $0.045 \rightarrow 0.341$ — a sign violation of $\dot{V} \leq 2\alpha V$. Collapse still occurs here, but it is rescued by a different argument (§4), not by the energy proof. The “exact, not approximate” reading applies only to the symmetric case.

3.2 The authoritative test: the full spectrum

$\alpha < 0$ on the symmetric part is *necessary but not sufficient* for strict contraction of the full system. The conservative, always-correct condition is

$$\max \operatorname{Re} \lambda(A) < 0 \quad \text{on the full } A \text{ (via eig, not eigvalsh)}. \quad (5)$$

A Bendixson bound gives $\operatorname{Re} \lambda(A) \leq \lambda_{\max}(A_s)$, so α bounds the real parts but does not by itself certify them: a rotation $A = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & -2 & 0 \end{pmatrix}$ has $\alpha = -1$ yet eigenvalues $\{-1, \pm 2i\}$ — a center that never collapses in its rotating plane. `collapse_certificate()` therefore reports *both* α (a conservative small-gain bound $\max \lambda(A_s) + \|A - A_s\|_2$) and $\max \operatorname{Re} \lambda(A)$ (the exact abscissa via eig, with a `full_contracting` flag).

Verification. The shipped test uses $A = -0.8I$, which is *symmetric* ($A = A_s$, hypothesis holds exactly, cross term identically zero). The certificate predicts contraction rate 0.8 and a rollout shows V decaying monotonically — confirming the theorem in precisely the special case where the proof is unconditionally valid; it does *not* exercise the non-normal case. (`collapse_certificate`, `lyapunov_value`, `src/cio_sde/collapse.py`.)

4 Provable region-wideners for non-normal A

The small-gain bound over-rejects strongly non-normal A , and $\max \operatorname{Re} \lambda(A) < 0$ alone is necessary-not-sufficient (transient growth). `stability_gates()` adds two *sufficient, provable* contraction certificates, each strictly wider than small-gain (margin $m \geq 0$):

1. **Numerical-range gate (monotone).** $\omega(A) = \lambda_{\max}(A_s) < -m \implies \|e^{tA}\|_2 \leq e^{\omega t}$ — a strict, no-transient contraction (matrix measure μ_2 ; Lohmiller–Slotine 1998 [8]). $\omega(A)$ is the rightmost point of the numerical range $W(A)$; since $\operatorname{spec}(A) \subset W(A)$, this gate *implies* the Lyapunov gate.
2. **Lyapunov gate (asymptotic, optimal metric).** For margin $m \geq 0$, $\exists P \succ 0 : (A + mI)^\top P + P(A + mI) \prec 0 \iff \max \operatorname{Re} \lambda(A) < -m$ (classical Lyapunov theorem [9]; $= \inf_T \mu_2(TAT^{-1}) < -m$). Certified by solving $(A + mI)^\top P + P(A + mI) = -I$ and checking $P \succ 0$ (with a relative ill-conditioning guard, so the boundary is never certified independent of solver warnings). It *accepts* the §3.1 counterexample $\begin{pmatrix} -1 & 3 \\ -3 & 0 \end{pmatrix}$ (Hurwitz at $-\frac{1}{2}$) that the energy proof and small-gain bound reject; $\sqrt{\kappa(P_0)}$ (the margin-0 solve) bounds the Euclidean transient $\sup_t \|e^{tA}\|$.

Honest scope. Sufficient, not necessary; they certify the *full* Jacobian’s contraction, not collapse-onto-manifold (the active/null alignment gap is separate, see §6). The *proven* transient constant is Crouzeix–Palencia (2017) [10]: $\|e^{tA}\| \leq 1 + \sqrt{2}$ when $W(A) \subset \text{LHP}$ ($\omega \leq 0$); the sharper constant 2 is Crouzeix’s still-open conjecture. These gates *extend the proven region of §3*; they do not make the system globally uncollapsible.

Verification. `tests/test_stability_gates.py` (14 tests): the $\begin{pmatrix} -1 & 3 \\ -3 & 0 \end{pmatrix}$ case (small-gain rejects, Lyapunov accepts, numerical-range flags the transient), a 400-matrix red-team finding *no* false-positive certificate, and matrix-exponential checks that the monotone ($e^{\omega t}$), Lyapunov ($\sqrt{\kappa(P)}$), and Crouzeix–Palencia $(1 + \sqrt{2})$ bounds each hold.

5 The collapse trigger Σ_0

Status: operational definition — not derived from Theorem 1.

Σ_0 fires when *all four* conditions hold simultaneously (a control singularity over a structureless field):

Condition	Meaning
$\ \nabla_x L\ < \varepsilon_g$	no optimization signal remains
$\text{rank}(J_f) < \rho n$	drift Jacobian has lost directional structure
Σ isotropically flat	uncertainty has no preferred direction
$\ \partial H / \partial u\ < \varepsilon_c$	control cannot distinguish actions

None of these four quantities is the spectral abscissa α of Theorem 1. They are an operational definition of “underdetermined” — a soft AND-gate $p = \min(p_\nabla, p_{\text{rank}}, p_{\text{aniso}}, p_{\text{ctrl}})$ (a Gödel t-norm) over four independent signs of degeneracy. The link between “the four conditions fire” and “ $\alpha < 0$ ” is a *modeling assumption*, not a proof. When triggered, Σ_0 projects the state onto the null eigenmodes of A_s ,

$$x^* = Px, \quad P = V_{\text{null}} V_{\text{null}}^\top, \quad (6)$$

the “42-state” (a colloquial name, no formal meaning).

Caveat. $x^* = Px$ is a true fixed point only when A is normal. For non-normal A the projection kills the symmetric part but the skew rotation leaves $\|Ax^*\|$ large, so x^* is not an equilibrium of the real flow. In the implementation the apparent “freeze” is produced by the integrator overwriting $x_{\text{next}} = x^*$ and discarding the diffusion term — a clamp, not an emergent attractor. (`SemanticCollapseOperator`.)

6 Lemma L2: the one-step anisotropy lift

The trigger’s flat leg fires when the eigenvalue coefficient of variation $a(\Sigma) = \text{std}(\lambda) / \text{mean}(\lambda) < \varepsilon_a$. The anti-collapse operator (§7) adds, in one step, $\Sigma^+ = \Sigma + bP$ with $b \geq 0$ and P a rank- k orthogonal projector. We prove a closed-form threshold above which one bump provably breaks flatness.

Lemma 1 (One-step anisotropy lift). *Let $\Sigma \succeq 0$ be real symmetric $d \times d$ ($d \geq 2$) with eigenvalues $\lambda_1, \dots, \lambda_d > 0$, mean $\mu > 0$, and population CoV $a = \sigma_{\text{pop}}/\mu < \varepsilon$ for some $\varepsilon \in (0, 1)$. Let P be an orthogonal projector of rank k , $1 \leq k \leq d-1$, whose range is spanned by k eigenvectors of Σ (alignment hypothesis). Then for $b > 0$,*

$$b \geq \Delta := \frac{(\varepsilon + a)\mu d}{\sqrt{k(d-k)} - \varepsilon k} \implies a(\Sigma + bP) \geq \varepsilon, \quad (7)$$

provided the denominator is positive, i.e. $\varepsilon < \sqrt{(d-k)/k}$ (true for every $1 \leq k \leq d-1$ when $\varepsilon \leq 0.05$ and $d \leq 400$).

Proof sketch. Because P aligns with eigenvectors of Σ , $\Sigma^+ = \Sigma + bP$ has the same eigenvectors and eigenvalues $\lambda_i^+ = \lambda_i + b$ on the bumped set B ($|B| = k$), $\lambda_j^+ = \lambda_j$ otherwise. By the law of total variance with weights k/d , $(d-k)/d$, adding the constant b to every element of B leaves the within-group variance invariant, so $(\sigma^+)^2 \geq \frac{k(d-k)}{d^2} (b - |\mu_B - \mu_U|)^2$. Bounding the pre-bump gap by the total spread, $|\mu_B - \mu_U| \leq a\mu d / \sqrt{k(d-k)}$, and using the exact post-bump mean $\mu^+ = \mu + (k/d)b$, the inequality $a(\Sigma^+) = \sigma^+/\mu^+ \geq \varepsilon$ reduces to $b(\sqrt{k(d-k)} - \varepsilon k)/d \geq (\varepsilon + a)\mu$, i.e. $b \geq \Delta$. Every step is an exact inequality, so Δ is a sufficient (conservative) threshold. $\square$

Consequence and scope. With $\varepsilon = \varepsilon_a$, one aligned bump of magnitude $b \geq \Delta$ lifts the anisotropy above threshold, so the trigger’s flat condition is false on the next step and the four-condition AND-gate cannot fire — the freeze is skipped that step. L2 is *unconditionally true* and machine-checked (`experiments/prove_l2_anisotropy_lift.py`; `tests/test_cio_sde.py::test_l2_anisotropy`). It does *not* by itself prove non-collapse: that needs the alignment hypothesis (proven for normal A , open for non-normal) and a proximity floor (an engineered hypothesis). L2 is the part that is unconditionally true.

7 The anti-collapse operator Σ_0^{-1}

Status: measured over distribution (900-run sweep, 100% prevented). L2 (§6) is the proven lemma toward a sufficiency theorem; the full theorem — non-normal A , all initial conditions — is the one open frontier.

Where Σ_0 projects *onto* the null subspace, Σ_0^{-1} injects energy *along* it:

$$dx = f dt + dW + \Sigma_0^{-1}, \quad \Sigma_0^{-1} = s \cdot p \cdot (V_{\text{null}} \xi), \quad (8)$$

with ξ standard normal, s the strength, and $p \in [0, 1]$ the collapse proximity — 0 far from the boundary (a no-op that costs nothing), rising toward 1 as ∇L , rank, anisotropy, and control sensitivity all approach their thresholds. The implementation injects noise in the null subspace and re-anisotropizes Σ (`AntiCollapseOperator.excite`); the proximity gate is the soft-AND of the four §5 signals.

Evidence (measured). `experiments/sigma0_regime_sweep.py` runs forced-collapse roll-outs with and without Σ_0^{-1} across an $\alpha \times$ non-normality $\times$ noise grid (fixed 3-dim-null Jacobian). Over the 900 **trials that genuinely collapse without protection**, Σ_0^{-1} suppressed collapse and re-excited the state in 100% (`data/sigma0_regime_sweep_report.json: collapse_prone_trials_total=900, headline_conditional_prevention_rate=1.0`). Honest caveat: in this construction the non-normal off-diagonal lifts the Jacobian’s effective rank, so the collapse-prone cells are the diagonal ones; a sufficiency *theorem* for all non-normal A remains future work.

Remark 1. On “persistent excitation.” The classical PE result (Anderson 1977 [6]) concerns estimator/parameter convergence under $\int \varphi \varphi^\top \geq \alpha I$. Here there is no estimator and no parameter being identified — only a state kept off a manifold. So Σ_0^{-1} is *inspired by* / *analogous to* persistent excitation; no PE condition is established.

8 The early-warning scalar (the “canary”)

Status: the eigenvalue readout is retired; the right signal — innovation surprise — is implemented (`src/cio_sde/surprise.py`).

Near a bifurcation the dominant eigenvalue flattens (critical slowing down [4, 5]). Eigenvalue-based readouts $p_{\text{unbounded}} = 1/|\text{Re } \lambda_{\max}(A_s)|$ and $p_{\text{gate}} = \text{clip}(1 - |\text{Re } \lambda_{\max}|/\varepsilon, 0, 1)$ were proposed but never drove the operator (the actual driver is the four-condition `proximity()`) and are formally retired.

The right canary, implemented. The correct early-warning is *surprise relative to uncertainty*: the Kalman normalized innovation squared $\text{NIS} = \nu^\top S^{-1} \nu$ with $\nu = y - C\hat{x}$, $S = C\Sigma C^\top + R$. $\text{NIS} \approx m$ means model and reality agree; $\text{NIS} \gg m$ means the model is overconfident and has drifted without knowing it — the standard innovation-consistency χ^2 test [7]. `forward_step` runs a Kalman predict/update cycle with process noise $Q = (g \cdot \text{dilation})^2 dt$, so smooth exploration stays consistent ($\text{NIS} \approx m$, silent) while the collapse snap / Σ_0^{-1} kick spikes NIS. `experiments/sigma0_horse_blinders.py` shows the regime the eigenvalue signal cannot: a low-coupling observer staying *calm while wrong* before the surprise spikes. The dangerous state is the quiet before the spook, not the spook.

9 Global structure: the attractor graph G

The system is multistable. Collect its attractors $\{A_1, \dots, A_k\}$, each with a basin $B_i = \{x_0 : \lim_{t \rightarrow \infty} \phi_t(x_0) \in A_i\}$. Coarse-graining to a graph $G = (V, E)$ gives an induced Markov process over basins, $P_{ij}(u) = \Pr(\pi(x_{t+1}) = A_j \mid \pi(x_t) = A_i, u_t)$, with partition map $\pi : \mathbb{R}^n \rightarrow V$. This is the standard Markov-state-model construction; it is genuinely Markov only under timescale separation (fast intra-basin relaxation vs. slow inter-basin hops). A boundary studded with *saddles* has stable manifolds — ridges one can ride with structured control (spin, the imaginary part $\Im \lambda$, is orthogonal to capture, the real part).

10 Demonstration on router data

Reproducible. The two driver scripts are committed and the numbers below are produced from committed code over the real conversation log (2678 turns): `python experiments/router_sigma0_encoder.py` and `python experiments/router_reservoir_G.py`.

Each turn is encoded as $x = [\text{novelty}, \text{self_repeat}, \text{echo}, \text{length}] \in [0, 1]^4$, with a local Jacobian fitted over a 10-turn sliding window by finite difference.

Encoder (2678 turns, 2673 with a fitted Jacobian)	Value
Mean Jacobian spectral radius ρ	1.064
Max ρ	27.93
Fraction of windows with $\rho > 1$	0.346
Reservoir G (echo-state, size 50, $\rho = 0.9$, ridge readout)	
One-step reconstruction MSE (held-out)	0.0097
Correlation dimension of reservoir trajectory	0.74

The mean spectral radius sitting just above 1 (a third of windows locally expanding) is the signature of a system perched near its stability boundary. The autonomous rollout (feeding the readout through the projection π onto $[0, 1]^4$) converges to a *low-dimensional fixed point* (correlation dimension ≈ 0.74) — the §3 prediction: absent external grounding the flow settles onto a single self-consistent state.

Honest deviation. The converged state is *not* the hand-sketched “parrot attractor”; the real log settles at high-novelty / low-echo / short-length. The qualitative claim (ungrounded recursion collapses onto a degenerate fixed point) is supported; the earlier specific numbers were not data-derived and are replaced by the produced ones. The source log is also partly synthetic test traffic, so the deliverable is the reproducible pipeline, not a population-level value.

11 Why this is a warning against ASI

Status: the one downstream claim worth keeping — as a machine-learning claim, on ML evidence, not as a consequence of the physics or of §10.

A system that optimizes against its own representations with no external anchor is the flow $\dot{x} = f(x)$ where f only ever sees x . The linearized certificate frames two degenerate fates absent outside contact: **collapse** onto a dead fixed point (the 42-state, mirrors agreeing with mirrors), or **divergence**. The only stable middle required an external bound. **Grounding is the safety mechanism.**

This does not need the certificate’s physics or the §10 numbers: it is the documented phenomenon of *model collapse* [12]. Dohmatob et al. [13] show synthetic data *changes the scaling law itself* — even a 1% synthetic fraction can truncate scaling — and Feng et al. [14] show that *verification* of synthesized samples prevents collapse: the published analogue of the external-grounding requirement. The phenomenon is closely related to reward hacking / specification gaming [15, 16].

Two honest qualifications. The strict “collapse or diverge, no third” dichotomy is the *linearized* framing, not a general ML theorem (real training also admits limit cycles and partial-information equilibria — better stated as “tends to degenerate or destabilize absent grounding”). And the in-repo demonstration, though now reproducible, runs on a partly-synthetic log, so the *argument* rests primarily on the published model-collapse literature.

12 Limitations and open problems

- **The non-collapse theorem is open.** L2 (§6) proves one aligned bump breaks flatness; a closed-form proof that Σ_0^{-1} *always* prevents collapse, for non-normal A and all initial conditions, is future work. The empirical support is a 900-run distribution, not a theorem.
- **Alignment for non-normal A .** The bump-basis-aligns-with-eig(Σ) hypothesis is proven only for normal A ; the non-normal cross-term is the same debt as Theorem 1.
- **Safety $\neq$ capability.** The certificate bounds collapse; it says nothing about whether the bounded system is useful.
- **Mechanization.** Theorem 1 and Lemma 1 are amenable to a Lean/Mathlib formalization (Weyl/interlacing + a scalar CoV inequality); we have machine-checked numerics but not yet a proof assistant.

13 Conclusion

A computable contraction certificate, two provable non-normal gates, a closed-form anisotropy-lift lemma, and a reproducible demonstration together support one disciplined claim: an ungrounded self-improving system has no benign limit, and external grounding is the mechanism that supplies one. We have been deliberate about the boundary between what is proven, what is measured, and what is heuristic — because a stability certificate is only worth as much as its honesty.

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Reproducibility. Code of record: `src/cio_sde/collapse.py` (`SemanticCollapseOperator`, `collapse_certificate`, `AntiCollapseOperator`, `stability_gates`), `src/cio_sde/surprise.py` (`SurpriseMonitor` + Kalman update). Tests: `tests/test_cio_sde.py` (34 passing; Theorem 1 and L2), `tests/test_stability_gates.py` (14 passing). Experiments: `sigma0_regime_sweep.py` (900-run sweep), `prove_l2_anisotropy_lift.py` (symbolic + numerical L2), `router_sigma0_encoder.py`, `router_reservoir_G.py` (§10), `sigma0_horse_blinders.py` (the canary).